

5.25 (a) Using the configuration on the right, we know

$$r_1^2 = d^2 - bd \cos \alpha + b^2/4, \quad r_2^2 = d^2 + bd \cos \alpha + b^2/4.$$

For a current at the origin in the $-\hat{z}$ direction, the vector potential at a distance r away from it is given by

$$\vec{A} = \hat{z} \frac{\mu_0 I_1}{2\pi} \log r.$$

Then, the interaction energy is

$$W_{12} = -I_2 a \hat{z} \cdot \hat{z} \frac{\mu_0 I_1}{2\pi} \log r_1 + I_2 a \hat{z} \cdot \hat{z} \frac{\mu_0 I_1}{2\pi} \log r_2 = \frac{\mu_0 I_1 I_2 a}{2\pi} \log \frac{r_2}{r_1}$$

$$= \frac{\mu_0 I_1 I_2 a}{4\pi} \log \left[\frac{4d^2 + b^2 + 4bd \cos \alpha}{4d^2 + b^2 - 4bd \cos \alpha} \right].$$

(b) The magnetic induction from an infinitely long current at a distance r away is

$$\vec{B} = -\frac{\mu_0 I_1}{2\pi r} \hat{z} \times \hat{r} \quad \text{The force is then given by}$$

$$\vec{F} = -I_2 a \hat{z} \times \left(-\frac{\mu_0 I_1}{2\pi r_1} \hat{z} \times \hat{r}_1 \right) + I_2 a \hat{z} \times \left(-\frac{\mu_0 I_1}{2\pi r_2} \hat{z} \times \hat{r}_2 \right)$$

$$= \frac{\mu_0 I_1 I_2 a}{2\pi} \left(\frac{\hat{r}_2}{r_2} - \frac{\hat{r}_1}{r_1} \right) = \frac{\mu_0 I_1 I_2 a}{2\pi} \left[\frac{\hat{x} \left(-\frac{b}{2} \sin \alpha \right) + \hat{y} \left(d + \frac{b}{2} \cos \alpha \right)}{d^2 + bd \cos \alpha + b^2/4} \right.$$

$$\left. - \frac{\hat{x} \left(\frac{b}{2} \sin \alpha \right) + \hat{y} \left(d - \frac{b}{2} \cos \alpha \right)}{d^2 - bd \cos \alpha + b^2/4} \right]$$

$$= \frac{\mu_0 I_1 I_2 a}{2\pi} \frac{-b \sin \alpha \left(d^2 + \frac{b^2}{4} \right) \hat{x} - b \cos \alpha \left(d^2 - \frac{b^2}{4} \right) \hat{y}}{\left(d^2 + b^2/4 \right)^2 - b^2 d^2 \cos^2 \alpha}$$

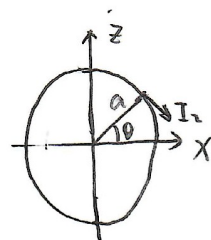
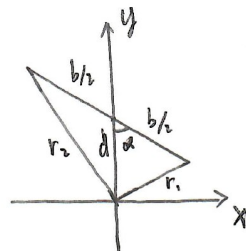
This is an attractive force. Interestingly, the y -component of the force is just $\partial W_{12} / \partial d$.

(c) Consider a ring originally in the xz plane, with center at the origin.

Points on the ring can be parameterized by angle θ . Then, for a point, we have

$$\vec{r} = (a \cos \theta, 0, a \sin \theta), \text{ and } \vec{I}_2 = -I_2 \hat{\theta} = I_2 (\sin \theta, 0, -\cos \theta). \text{ Now, rotate}$$

this ring around $\hat{z}$ -axis by an angle of $\frac{\pi}{2} - \alpha$, then



$$\vec{r} = (a \cos \theta \sin \alpha, -a \cos \theta \cos \alpha, a \sin \theta), \quad \vec{I}_2 = I_2 (\sin \theta \cos \alpha, -\sin \theta \sin \alpha, -\cos \theta)$$

Finally, move the center of ring along y-axis by a distance of d , then

$$\vec{r} = (a \cos \theta \sin \alpha, d - a \cos \theta \cos \alpha, a \sin \theta), \text{ while } \vec{I}_2 \text{ stays the same.}$$

Following part (a), the interaction energy is

$$W_{12} = \int \vec{I}_1 \cdot \vec{A} \, dl = \int (-I_2 \cos \theta \hat{z}) \cdot \left(\hat{z} \frac{\mu_0 I_1}{2\pi} \log r \right) dl = -\frac{\mu_0 I_1 I_2 a}{2\pi} \int_{-\pi}^{\pi} \cos \theta \log r \, d\theta,$$

where r is the distance from a point on the ring to z-axis,

$$r^2 = (a \cos \theta \sin \alpha)^2 + (d - a \cos \theta \cos \alpha)^2 = |d - a \cos \theta e^{i\alpha}|^2,$$

$$\text{and } \log r = \log |d - a \cos \theta e^{i\alpha}| = \operatorname{Re} [\log (d - a \cos \theta e^{i\alpha})]$$

$$\text{Therefore, } W_{12} = -\frac{\mu_0 I_1 I_2 a}{2\pi} \operatorname{Re} \left[\int_{-\pi}^{\pi} \cos \theta \log (d - a \cos \theta e^{i\alpha}) \, d\theta \right]$$

If you ask Google to calculate $\text{Integrate} [\operatorname{Ln}[1 - px] \cos[x], \{x, -\pi, \pi\}]$ (Mathematica grammar), it will tell you the result is $-2\pi(1 - \sqrt{1-p^2})/p$, which leads to

$$\begin{aligned} W_{12} &= -\frac{\mu_0 I_1 I_2 a}{2\pi} \operatorname{Re} \left[-2\pi \left(1 - \sqrt{1 - \frac{a^2}{d^2} e^{2i\alpha}} \right) / \frac{a}{d} e^{i\alpha} \right] \\ &= \mu_0 I_1 I_2 d \operatorname{Re} \left[e^{-i\alpha} - \sqrt{e^{-2i\alpha} - a^2/d^2} \right] \end{aligned}$$

(d) In the circular case, $\sqrt{e^{-2i\alpha} - a^2/d^2} = e^{-i\alpha} \left(1 - \frac{a^2}{2d^2} e^{2i\alpha} \right) = e^{-i\alpha} - \frac{a^2}{2d^2} e^{i\alpha}$. Then

$$W_{12} = \mu_0 I_1 I_2 d \operatorname{Re} \left[\frac{a^2}{2d^2} e^{i\alpha} \right] = \frac{\mu_0 I_1}{2\pi d} \cdot \pi a^2 I_2 \cos \alpha = \vec{m} \cdot \vec{B}.$$

In the rectangular case,

$$\log \left(d^2 \pm db \cos \alpha + \frac{b^2}{4} \right) = \log d^2 \pm \frac{b}{d} \cos \alpha + o\left(\frac{b}{d}\right),$$

$$\text{and } W_{12} = \frac{\mu_0 I_1 I_2 a}{4\pi} \frac{2b}{d} \cos \alpha = \frac{\mu_0 I_1}{2\pi d} \cdot I_2 ab \cos \alpha = \vec{m} \cdot \vec{B}.$$